

Imprecision in inferential models: why, what, and how

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Statisticians want numerical measures of the degree to which data support hypotheses. —Hacking

- A/the goal of statistical inference is quantifying uncertainty about unknowns based on data, models, etc
- Hacking says we want “numerical measures of...”
- Gut reaction: “numerical measures of...” = “probability”
- My claim: *probability doesn't provide reliable uncertainty quantification in this context*
- I've been advocating for inferential models (IMs) that take the mathematical form of an *imprecise probability*

- FAQs related to these claims:
 - *why imprecision?*
 - *what kind of imprecision?*
 - *how to achieve this imprecision?*
- This talk aims to answer these questions
- Rough summary:
 - why*: precision pushes likelihood too far → unreliability
 - what*: possibility theory is the right² mode of imprecision
 - how*: imprecise-probability-to-possibility transform/p-values
 - why*: some interesting properties

²It's also the simplest mode, an added benefit

- Data $Y \in \mathbb{Y}$, observed value y
- Model $\{P_\theta : \theta \in \mathbb{T}\}$, e.g., $\text{Bin}(n, \theta)$
- Uncertain true value is Θ , generic values denoted by θ
- Assume no prior info about Θ is available
- Data + model $\implies$ likelihood: $\theta \mapsto L_y(\theta)$
- **Goal:** *quantify uncertainty about Θ , given $Y = y$*
 - Hacking says we want a mapping $H \mapsto Q_y(H)$
 - Imagine a y -dependent “probability” Q_y for Θ
- Often L_y drives this probabilistic UQ...

The function of the θ 's... is not however a probability and does not obey the laws of probability; it involves no differential element $d\theta_1 d\theta_2 d\theta_3 \dots$; it does none the less afford a rational basis for preferring some values of $\theta \dots$ to others. —Fisher

- Fisher's warnings "likelihood $\neq$ probability" are familiar
- But we don't take this seriously³
- Two-part take-away:⁴ for a hypothesis " $\Theta \in H$ "
 - "no differential element" $\implies$ size of H has no bearing
 - only *relative likelihood* values on H are relevant

³To be fair, even Fisher himself didn't take this seriously...

⁴M., arXiv:2303.17425

Pushing likelihood too far, cont.

- Default prior Bayes:
 - neither prior nor likelihood has meaningful differential element
 - how can posterior's differential element be meaningful?
- e.g., $\{Y \sim N(\Theta, 1), y = 7\} \rightsquigarrow Q_y = N(7, 1)$
 - two hypotheses $H = (6.5, 7]$ and $H' = (-\infty, 7]$
 - clearly, $Q_y(H') \gg Q_y(H)$
 - but is H' really more supported by data than H ?

To allow the size of the crowd of hypotheses... to influence the value of the [plausibility] assigned to any particular hypothesis, would be like weakening one's praise for the chief actors in a play on the ground that a large number of supers were also allowed to cross the stage. —Shackle

Imprecision: *why?*

- Pushing likelihood too far has consequences
- In particular, the following would be problematic:
 - for some false hypotheses $H \not\in \Theta$
 - the rv $Y \mapsto Q_Y(H)$ tends to be large under P_Θ
- *False confidence*⁵ is a particular brand of unreliability, and all probabilistic UQ is afflicted by it

False confidence theorem.

Let Q_Y be *any* data-dependent probability for Θ . For any thresholds (τ, ρ) , there exists hypotheses $H \subset \mathbb{T}$ such that

$$H \not\in \Theta \quad \text{and} \quad P_\Theta\{Q_Y(H) > \tau\} > \rho.$$

⁵Balch, M., & Ferson 2019, *Proc Roy Soc A*, arXiv:1706.08565

Imprecision: *what?*

- Reliable UQ requires something different from probability
- What else is there? *Imprecise probability*
 - “imprecise” $\neq$ “inaccurate”
 - it’s limiting the precision of statements made because being any more precise risks being unreliable
- Roughly, imprecise probability = *sets* of probabilities
- Inferential model (IM):
 - given data y , model, etc
 - to every $H \subset \mathbb{T}$, assign a pair of numbers

$$\underline{\Pi}_y(H) \quad \text{and} \quad \overline{\Pi}_y(H)$$

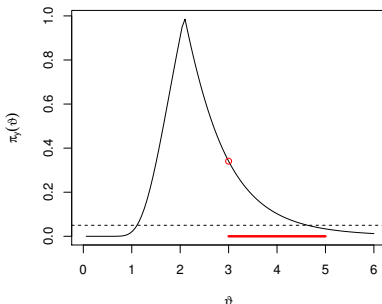
- y -dependent measures of support for and plausibility of H

- *Possibility theory*⁶ is the simplest imprecise probability
- Similar to probability theory
 - determined by a “density function”
 - but different calculus — optimization replaces integration
 - no differential element!
- Specifically:
 - possibility contour $\theta \mapsto \pi_y(\theta)$ with $\sup_{\theta} \pi_y(\theta) = 1$
 - define $\overline{\Pi}_y(H) = \sup_{\theta \in H} \pi_y(\theta)$
 - $\underline{\Pi}_y(H) = 1 - \overline{\Pi}_y(H^c)$
- What makes the IM output meaningful is its properties

⁶Didier Dubois is a leading authority on possibility theory; he did a “postdoc” with Purdue electrical engineering in 1979

Imprecision: *what?* – cont.

- Possibility contour plot
- Hypothesis $H = [3, 5]$
- $\bar{\Pi}_Y(H)$ via optimization
 - “most possible member”
 - no differential element
 - size of H is irrelevant
- Level set defines interval of “sufficiently possible” θ s



Imprecision: *how?*

- Special case of a much more general construction⁷
- Workhorse: *imprecise-probability-to-possibility transform*⁸
- In the “no-prior-info” case, this takes a simple form:
 - let $R(y, \theta) = L_y(\theta) / L_y(\hat{\theta}_y)$ be relative likelihood
 - define a possibility contour

$$\pi_y(\theta) = P_\theta\{R(Y, \theta) \leq R(y, \theta)\}, \quad \theta \in \mathbb{T}$$

- then $\bar{\Pi}_y(H) = \sup_{\theta \in H} \pi_y(\theta)$, etc.
- Not unfamiliar:⁹
 - contour is just the p-value for a likelihood ratio test
 - but classical theory doesn't go all the way to possibility

⁷M. arXiv:2211.14567

⁸e.g., Hose 2022 PhD thesis (Univ of Stuttgart)

⁹See, also, M. arXiv:1203.6665 and M. arXiv:1511.06733

What makes the IM output meaningful is its properties. —M

Validity theorem.

The new IM construction above is (strongly) valid, i.e.,

$$\sup_{\Theta} P_{\Theta} \{ \pi_Y(\Theta) \leq \alpha \} \leq \alpha, \quad \alpha \in [0, 1]$$

- Most basic property, familiar for p-values
- While the result itself isn't new/surprising, there are striking consequences wrt possibilistic UQ...

Corollary.

IM-based procedures are reliable in a frequentist sense:

- 1 the test “reject $\Theta \in H$ iff $\bar{\Pi}_Y(H) \leq \alpha$ ” satisfies

$$\sup_{\Theta \in H} P_{\Theta}\{\bar{\Pi}_Y(H) \leq \alpha\} \leq \alpha, \quad \alpha \in [0, 1], \quad H \subseteq \mathbb{T}$$

- 2 the region $C_{\alpha}(y) = \{\theta : \pi_y(\theta) > \alpha\}$ satisfies

$$\sup_{\Theta} P_{\Theta}\{C_{\alpha}(Y) \not\ni \Theta\} \leq \alpha, \quad \alpha \in [0, 1]$$

- These are the usual frequentist properties:
 - Type I error rate control
 - Nominal coverage probability

Corollary.

The error rate control above is *uniform in H* , i.e.,

$$\sup_{\Theta} P_{\Theta} \left[\bigcup_{H: H \ni \Theta} \{Y : \bar{\Pi}_Y(H) \leq \alpha\} \right] \leq \alpha$$

- The goal isn't just “test one H and done”
- Often the goal is to *probe*¹⁰
 - e.g., if null is rejected, find other H 's that are supported
 - follow-ups depend on data, so H -wise control isn't enough
- Corollary provides error rate control under probing¹¹

¹⁰Related to Deborah Mayo's “severe testing”

¹¹Cella and M., arXiv:2304.05740

- Imprecise prob corresponds to sets of ordinary probabilities
- This (non-empty) set is called the *credal set*

$$\mathcal{C}(\overline{\Pi}_y) = \{Q_y \in \text{probs}(\mathbb{T}) : Q_y(\cdot) \leq \overline{\Pi}_y(\cdot)\}$$

- For possibility measures $\overline{\Pi}_y$, there's a characterization:

$$Q_y \in \mathcal{C}(\overline{\Pi}_y) \iff Q_y(\underbrace{\{\theta : \pi_y(\theta) > \alpha\}}_{100(1-\alpha)\% \text{ pl region}}) \geq 1 - \alpha$$

- i.e., elements of $\mathcal{C}(\overline{\Pi}_y)$ are confidence distributions
- More can be said, with additional structure

- Suppose the model has a group transformation structure
- e.g., location–scale models
- With such structure, all the following methods give the same solution for probabilistic UQ:
 - default-prior Bayes
 - (generalized) fiducial
 - ...

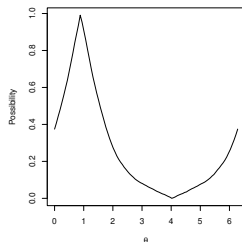
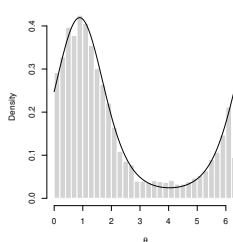
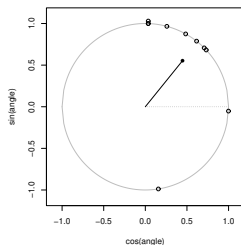
Theorem (M., arXiv:2303.08630).

Under the structure above, let Q_y^* be the fiducial solution and $\overline{\Pi}_y$ the possibilistic IM. Then $Q_y^* \in \mathcal{C}(\overline{\Pi}_y)$ and, moreover,

$$Q_y^*(\{\theta : \pi_y(\theta) > \alpha\}) = 1 - \alpha, \quad \text{all } \alpha \in [0, 1]$$

Example

- Directional data, i.e., angles or points on a circle
- Simple von Mises model with unknown mean direction Θ
- Group structure: Θ plays the role of a rotation
- Plots show data,¹² fiducial density, and IM contour¹³



¹²From Example 1 in Mardia's *Directional Statistics*

¹³Details in M., arXiv:2303.08630

- Imprecision in IMs: why, what, and how
- More specifically,
 - why $\rightarrow$ reliability guarantees, no false confidence
 - what $\rightarrow$ possibility theory
 - how $\rightarrow$ imprecise-prob-to-poss transform
- Possibilistic IM has very strong properties
- New connection between IMs and fiducial
- Other consequences of this connection (re: false confidence) are presented in the paper¹⁴

¹⁴M., arXiv:2303.08630



ST 790 (001) Fall 2022 Advanced Special Topics

Imprecise-Probabilistic Foundations of Statistics & Data Science

`https://wordpress-courses2223.wolfware.ncsu.edu/
st-790-001-fall-2022/`

- Links to papers, talks, etc. can be found on my website:

`www4.stat.ncsu.edu/~rmartin/`

- If you have questions, etc, email me at `rgmarti3@ncsu.edu`

Thanks for your attention!